Low Level Design (LLD)

Project Title

Design & Build a RAG-Based Customer Support Assistant

Modules

- Document Processing Module
- Chunking Module
- Embedding Module
- Vector Storage Module
- Retrieval Module
- Query Processing Module
- Graph Execution Module
- HITL Module

Data Structures

Document Object, Text Chunks, Embedding Vectors, Query Input, Response Output, Graph State.

Workflow Logic

Input Node -> Processing Node -> Output Node

Conditional Routing

Matched query -> Answer; Low confidence -> Escalate; No context -> Manual support.

Error Handling

Missing PDF, Empty Query, API Failure, No Relevant Result.